THE QUESTION: COE or ERP: which one actually keeps Singapore's roads moving?

Roads got about a quarter more crowded. Speeds held.

17 October 2026 · ~9 min read

ECONOMICS

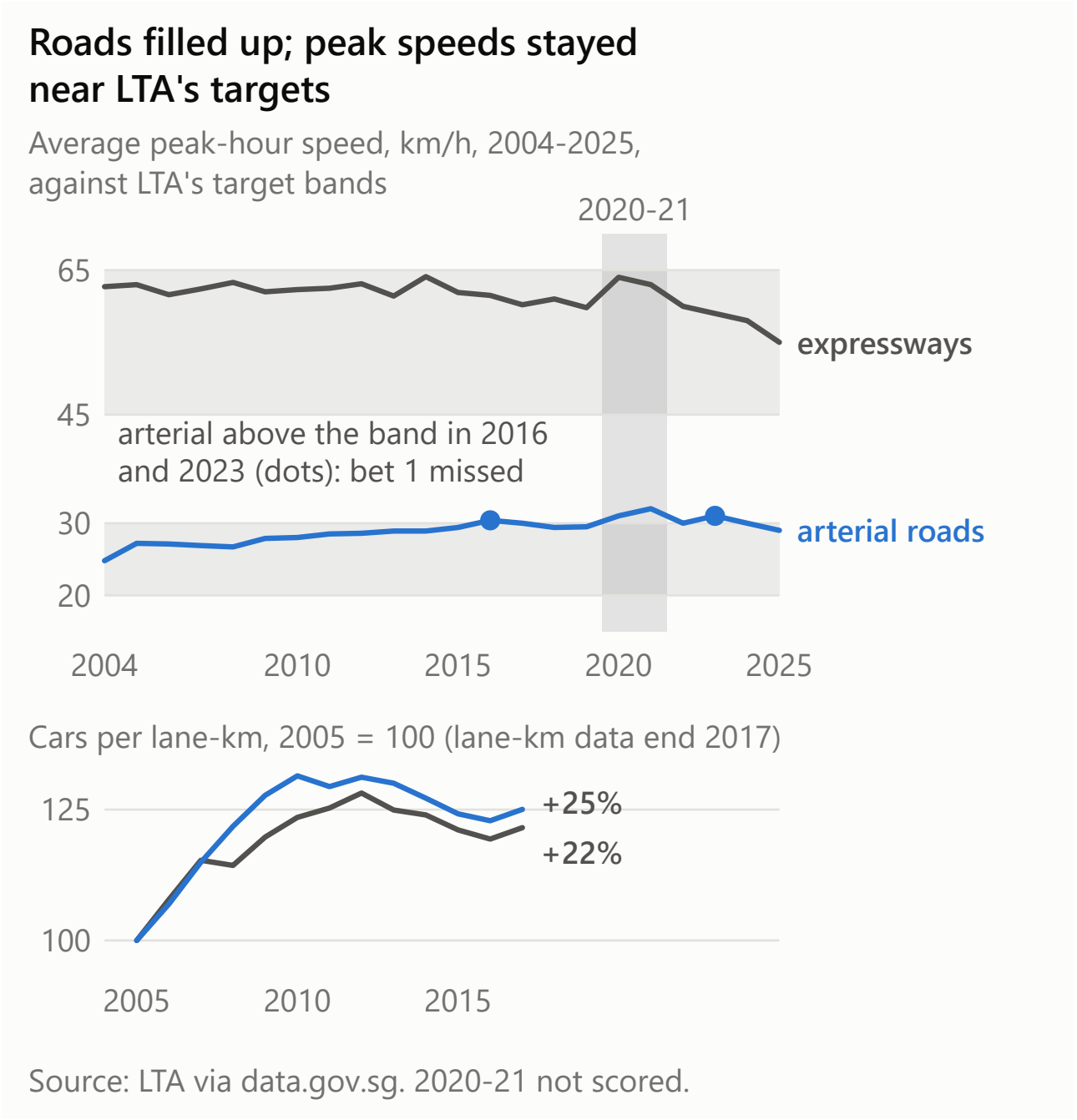
PRE-REGISTERED 26 September 2026

TESTS 1 passed, 2 failed, 1 inconclusive of 4 RE-CHECK DUE 17 October 2027

Repository →

Re-check due: the date this finding is re-tested against new data. How the scorecard works

Singapore prices owning a car with COE and using a busy road with ERP. From 2005 to 2017 the roads got about a quarter more crowded per lane and peak-hour speeds barely moved. Four predictions were sealed before the data was opened; one held, and the record could not say which instrument deserves the credit.



Source: LTA, Average Speed During Peak Hours, 2004-2025, and Annual Motor Vehicle Population by Vehicle Type and Length of Roads (Lane-Kilometres), 2005-2017, via data.gov.sg; target bands from the Ministry of Transport.

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Verdict: the record cannot tell COE and ERP apart.
Key number: from 2005 to 2017, cars per lane of road rose by 22 to 25 per cent; peak-hour speeds barely moved.
My confidence going in: 1.9 of 4 predictions expected to hold.
Scored: 1 of 4 held.
What would change it: a year-by-year record of how hard ERP was charging, or several years of peak speeds below LTA's target band.

Singapore charges for a car twice. The Certificate of Entitlement charges for owning one, once. Electronic Road Pricing charges for using a busy road at a busy hour, every time. Both are said to keep traffic moving. In September I wrote down four bets on what the record would show, with how sure I was of each, and put them on an open question page before looking at the data. One held. The roads got more crowded and stayed about as fast, which is what ERP promises, but the record cannot say whether ERP is the reason.

The measure throughout is weekday peak-hour road speed, as the Land Transport Authority (LTA) publishes it.

Each side has a strong case

For COE. A car that is never bought is never driven, parked or left idling at a junction. The quota is a hard ceiling fixed in advance, so car numbers cannot run away, and it deals with every cost of a car at once, not only congestion.

For ERP. Congestion is about how many cars are in one place at one time. A charge on the jammed road at the jammed hour lands where the harm is done, and it corrects itself: LTA raises it when speeds fall below a target band and cuts it when they rise above. A fee paid once changes behaviour once. A charge on every trip changes it every morning.

Theory leans one way, then hedges

In 1920 Arthur Pigou argued that the right charge for anything that harms other people is the size of the harm. That is what ERP tries to be. William Vickrey later worked out why the charge belongs on the peak, and Singapore's 1975 Area Licensing Scheme was the first in the world to charge for entering a congested city centre. In 1974 Martin Weitzman asked when a cap on quantity beats a price: when harm climbs steeply past some point, a cap is safer, because a price set slightly wrong lets the harm run away. Congestion is that kind of harm, which on paper favours COE. But ERP is not a fixed price.

It is reviewed against a speed target, and a price that keeps chasing a target starts to behave like a cap.

Pushed to the extreme, they predict different roads

These are thought experiments. Singapore runs both.

Extreme COE is a buffet. A huge fee at the door, every plate free after. Few cars, but the 8am trip costs the owner nothing extra, so the few cars still crowd the same roads at the same hour. If this were the whole story, speeds would fall as more cars were packed onto each lane.

Extreme ERP is surge pricing. Anyone can own a car, but a jammed road at a jammed hour costs more until it moves. The roads move. If this were the whole story, speeds would stay inside the target band however crowded the roads got.

My four bets

1. **The band holds** (35%). Every year, average peak speed sits inside LTA's target: 45 to 65 km/h on expressways, 20 to 30 km/h on arterial roads.
2. **Speed ignores crowding** (65%). Once road space is allowed for, peak speed barely moves as the number of cars per lane-kilometre changes.
3. **A dearer door does not cut driving** (20%). In the 7 years with the highest COE premium, the average car was driven no more than 3 per cent less than in the 7 cheapest.
4. **The price adds nothing beyond the count** (70%). Once the number of cars is known, the premium says nothing more about peak speed.

The percentages are how sure I was that each would hold. Together they put the expected number holding at 1.9 of 4. The exact wording is in the sealed thesis.

The roads filled up and speeds held

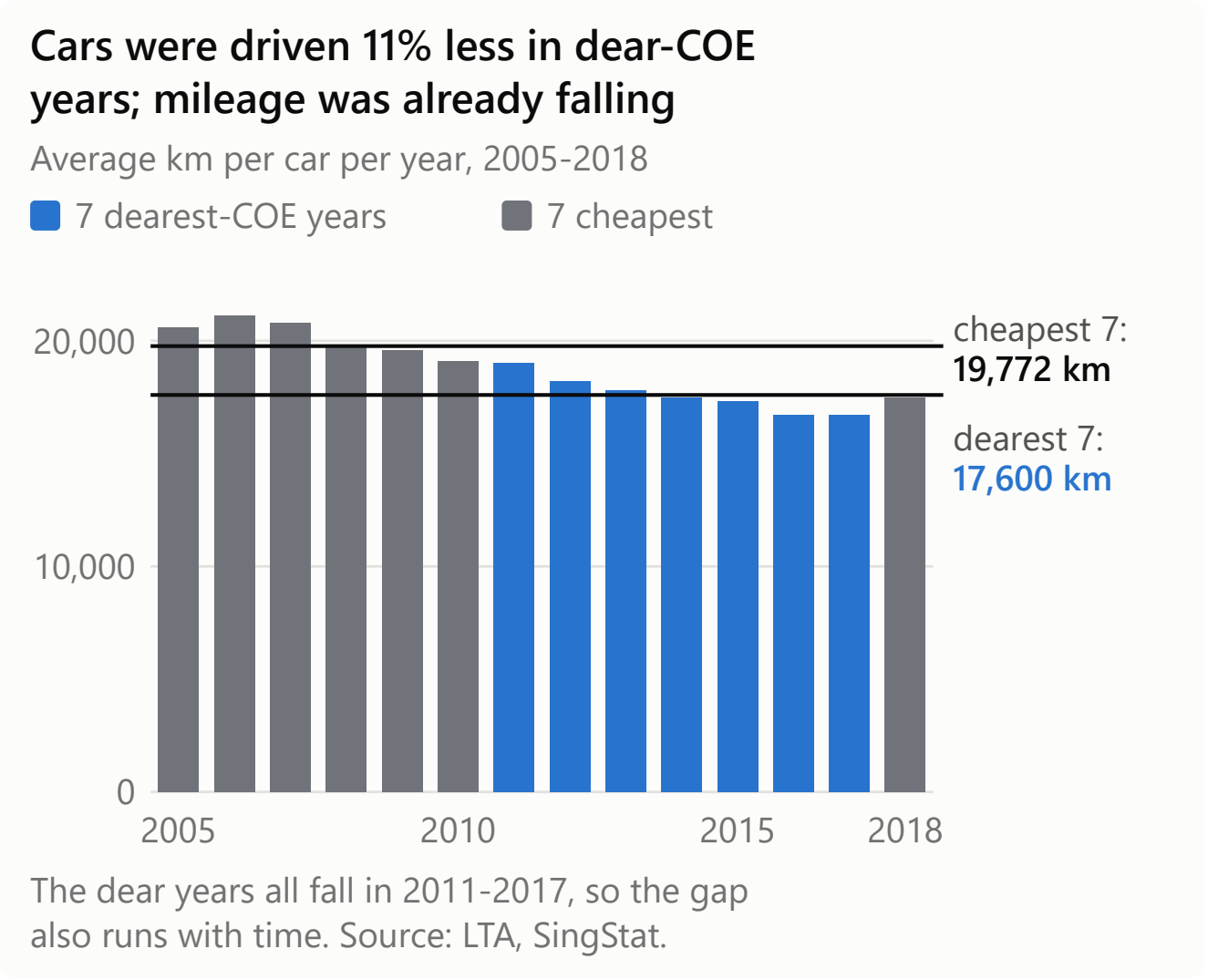
Between 2005 and 2017 the number of cars per lane-kilometre rose by 22 per cent on expressways and 25 per cent on arterial roads. Under the buffet, that would have shown up as slower traffic. It did not. On expressways, peak speed did not move with crowding at all. On arterial roads it drifted up, not down, while crowding rose. Bet 2 held on both road types.

That is the result ERP would produce if it works as designed. It is also the result new rail lines, wider roads or a steady shift in when people travel would produce. This is the thermostat problem: a thermostat that works leaves the room at the same temperature whatever the weather, and a flat temperature record looks exactly like a heater that did nothing on a mild day. A flat speed record cannot tell a working controller from calm conditions.

Where I was wrong

Bet 1 broke on the fast side. Expressways stayed inside 45 to 65 km/h. Arterial roads did not: 30.4 km/h in 2016 and 31 km/h in 2023, both above the top of the band. No year, on either road type, fell below its band. The buffet predicts roads slower than the band; this miss is the opposite, arterial traffic moving faster than LTA's target.

Bet 3 missed, and I was right for the wrong reason. I gave it only 20 per cent, because the mileage data counts Grab and other hire cars with private cars, and I expected their growth in the cheaper-COE years to make those years look like heavier driving. The high-premium years did come out lower: 17,600 km per car against 19,772, 11 per cent less, well past the 3 per cent line. But the hire cars grew in the wrong years for my story. Hire and rental cars rose by 54,736 over the seven high-premium years and by only 5,591 over the seven cheap ones. That pushed toward the bet holding, and it missed anyway.



Source: LTA, Average Annual Kilometres Travelled Per Vehicle, via data.gov.sg, 2005-2018; SingStat, Motor Vehicle Quota, Quota Premium And Prevailing Quota Premium, Monthly.

What does line up with the split is time. All seven dear years fall between 2011 and 2017, and km per car fell in every year from 2007 to 2016, including years when premiums were low. The data cannot separate “a pricier door cut driving” from “driving per car was drifting down anyway”.

Bet 4 came out mixed. On expressways, once the number of cars was known, the COE premium added nothing. On arterial roads it added a little, and that little was clear of zero. My own rule, written in advance, calls that neither a pass nor a fail, so it does not count as holding.

The tally. 1 of 4 held against the 1.9 I expected. The Brier score, which measures how far each confidence sat from what happened, was 0.194. A forecaster who always says 50 per cent scores 0.25, and lower is better, but four predictions is far too few to judge anyone’s calibration.

What would change my mind

- Peak speeds falling below the band for several years on either road type: the buffet showing through.
- Peak speed dropping at least one for one with cars per lane, consistently rather than in one odd year.
- A year-by-year record of how hard ERP was charging. Without it, half of the case for COE, that speeds tracked car numbers with ERP adding little, could not be tested at all.

The verdict

The record cannot tell the two apart.

A pass for the ERP reading is weak evidence and a COE-like result is strong evidence, because the design leans toward the ERP reading.

What this does not explain

Speed is one part of what a car costs everyone else. Emissions, road accidents and the land given over to roads and parking are the others, and none of them was measured here. A verdict on speed is not a verdict on the whole cost of a car. Nor does it say how hard ERP was pushing in any given year: no dated record of ERP rate changes before mid-2018 could be found.

+ **Method notes**

Sources

All data retrieved 26 September 2026 (Singapore time).

- LTA, Average Speed During Peak Hours, data.gov.sg, d_26f6afadf2f86b2004f9a1e28f5564cc, 2004-2025.
- LTA, Average Annual Kilometres Travelled Per Vehicle, data.gov.sg, d_bdc4c6434e47b055de4b5f2fde10c1af, 2005-2018.
- LTA, Annual Motor Vehicle Population by Vehicle Type, data.gov.sg, d_2873f3b1b2a836103f51f696350b98fa, 2005-2024; 2025 from LTA, Annual Vehicle Statistics 2025, MVP01-1.
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- LTA, Length of Roads (Lane-Kilometres), data.gov.sg, d_8415afe86e594bdc18f0f04a71d5f210, 2005-2017.
- Ministry of Transport, "How ERP works as a speed booster" (the target bands).
- LTA news releases on ERP, March 2020 to August 2026 (the 2020 suspension).
- W. Theseira (2020), "Congestion Control in Singapore", ITF/OECD discussion paper.
- A. C. Pigou (1920), *The Economics of Welfare*; W. S. Vickrey (1969), "Congestion theory and transport investment", *American Economic Review*; M. L. Weitzman (1974), "Prices vs. Quantities", *Review of Economic Studies*.

Code, data, the sealed thesis and every number above: erp/ in coe-analysis.

A historical reading of public data, in past tense throughout. It is not advice, and nothing here is tailored to any reader's own car or travel.

For economists: data, method and code

Data	LTA, Average Speed During Peak Hours, 2004-2025, and Annual Motor Vehicle Population by Vehicle Type and Length of Roads (Lane-Kilometres), 2005-2017, via data.gov.sg; target bands from the Ministry of Transport. LTA, Average Annual Kilometres Travelled Per Vehicle, via data.gov.sg, 2005-2018; SingStat, Motor Vehicle Quota, Quota Premium And Prevailing Quota Premium, Monthly.
Method	For each road class separately, OLS of log(annual peak speed) on log(cars per lane-km of that road class), over 2005-2017. OLS of log(speed) on log(car population), then again with log(annual premium) added.
Sealed	26 September 2026. 6d2a345 (erp: SEAL THESIS.md)

Code coe-analysis/erp

Rerun `rm -rf erp/out erp/figs && bash erp/run_all.sh`

Check erp/CHECKSUMS.md5; GitHub Actions

Changelog: 28 October 2026: Notes and Sources added under the figures; no number changed.

Questions and counter-arguments on this article go to Jacob's inbox.

Economics

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Questions and counter-arguments

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